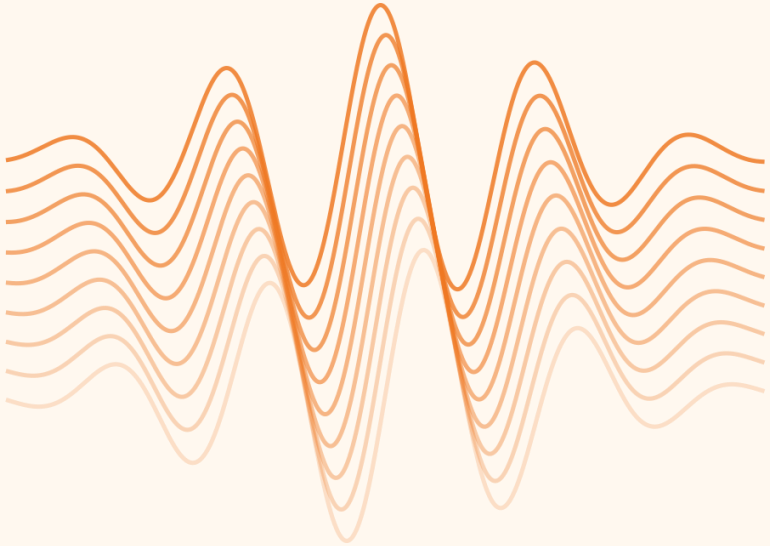
CalmGesture

Deep learning for gesture-controlled sound



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Problem statement & motivation

A touch-free way to explore sound

Conventional controls require physical input

Rule-based mappings offer limited recognition

Movement varies across users

Objective

Recognize four hand movements with deep learning and map them to gentle tones.

Exploratory prototype for autism-support research. Clinical benefit remains untested.

Dataset & data pipeline

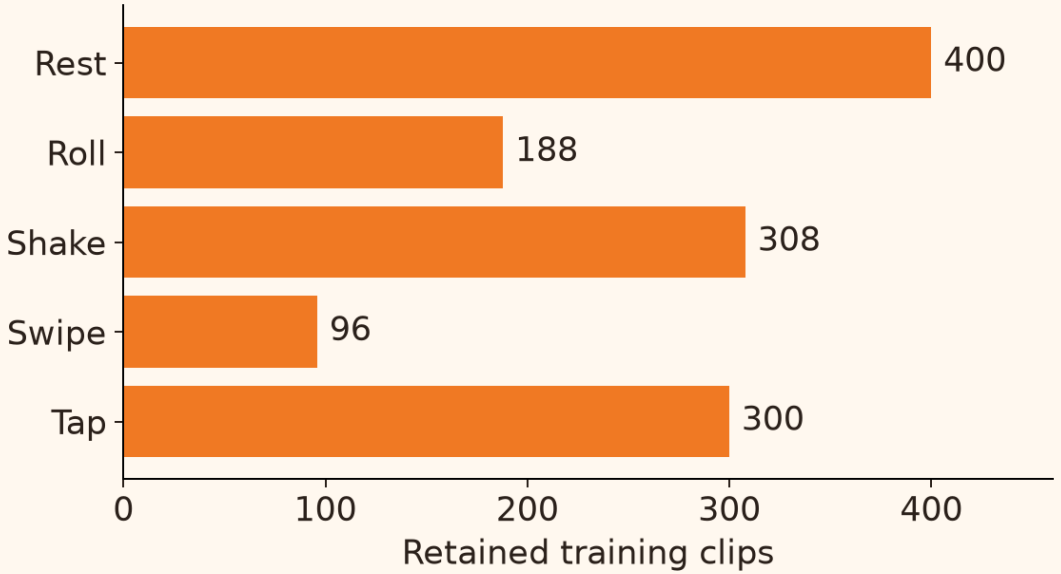
3,000 clips

Jester subset, five grouped classes

Candidate split: 66.7 / 16.7 / 16.7%

Retained: 1,292 / 500 / 500

Train / Validation / Test



21 landmarks × 3 coordinates × 30 frames

Motion normalization, missing-frame filling, resampling
No augmentation or class balancing in this run

System architecture & model choice

Baseline: MLP on motion summaries



Temporal filters retain movement order

32,901 parameters support lightweight inference

Best validation macro-F1: 76.98%

Training methodology & hyperparameters

Loss / optimizer	**Cross-entropy / Adam**
Learning rate / batch	**0.001 / 32**
Regularization	**Dropout 0.2, weight decay 0**
Training schedule	**50 maximum epochs, patience 10**
Checkpoint / seed	**Epoch 13 / seed 42**
Infrastructure	**Apple M1 CPU, four threads**

8.04 seconds for this model training run. Data preparation excluded.

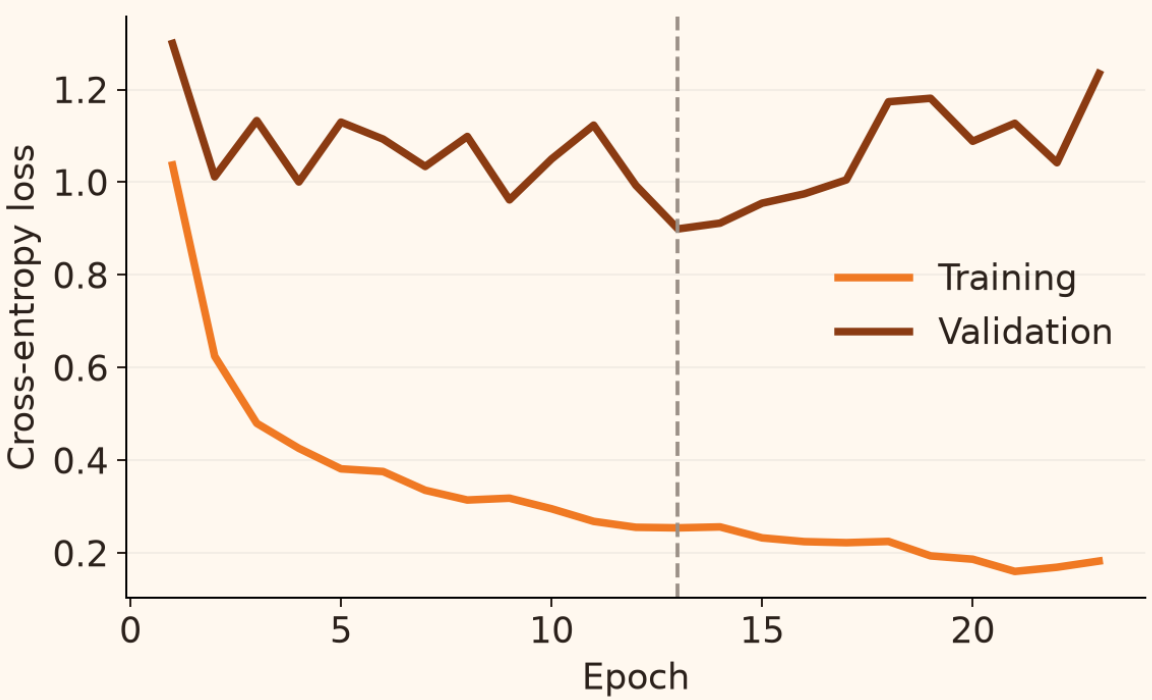
Quantitative results

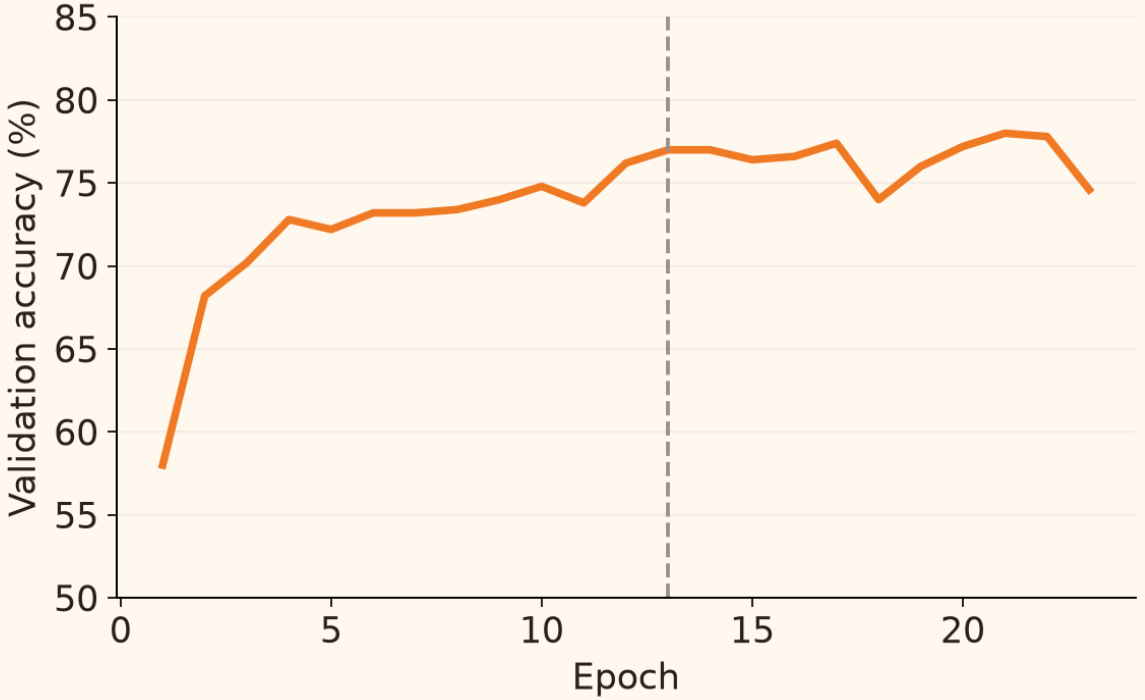
Model	Val. macro-F1	Test accuracy	Test macro-F1
MLP baseline	92.20%	93.60%	93.80%
NumPy CNN	92.97%	91.00%	91.60%
1D-CNN	**93.98%**	**96.80%**	**96.92%**
GRU	89.67%	93.00%	92.84%

1D-CNN improves baseline accuracy by 3.2 points

SOTA comparison unavailable under this five-class protocol.
Full Jester benchmarks use different inputs and evaluation settings.

Training dynamics





Best checkpoint: epoch 13

Validation loss rises as training loss falls. Early stopping limits overfitting.
Accuracy curve shows validation only. Training accuracy was not logged.

Qualitative results & error analysis

Success: shaking / shaking

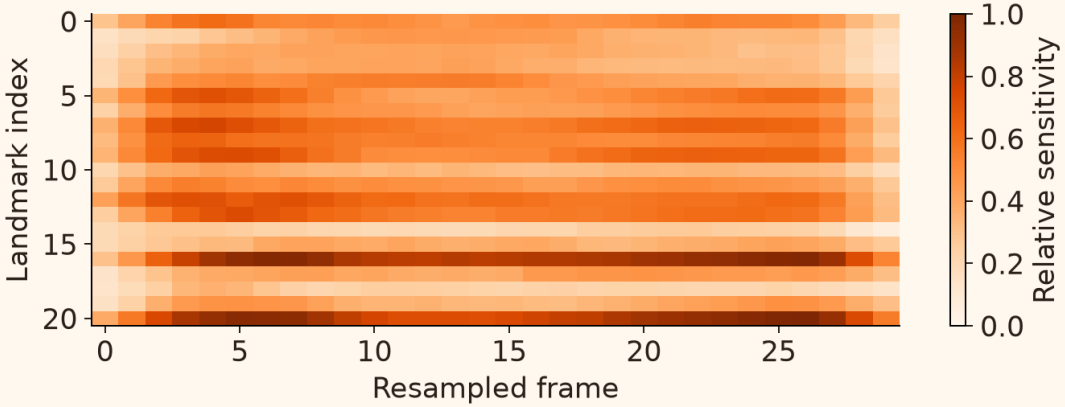


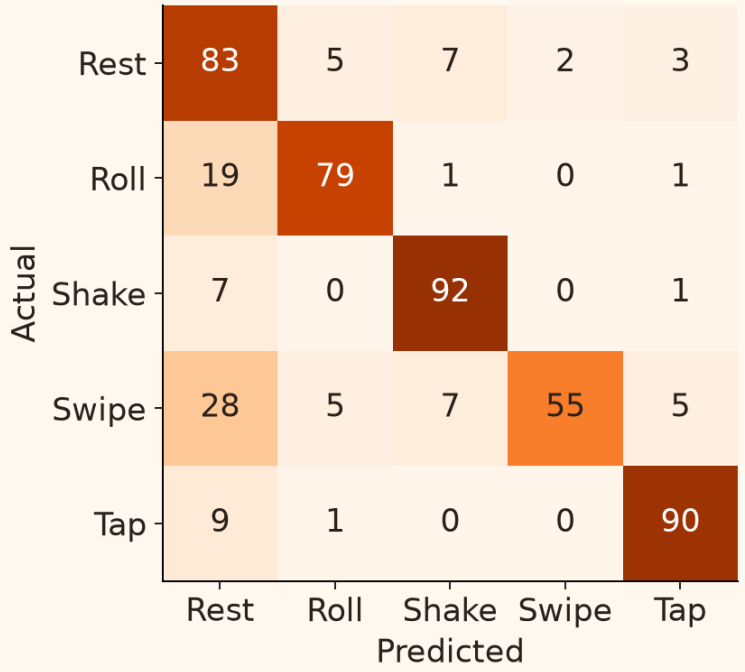
Clip 118649 · actual / predicted

Failure: swiping / background



Clip 23283 · actual / predicted





Swiping recall: 87%

28 swipes predicted as rest

Input-gradient sensitivity, not causal proof

Discussion & key takeaways

What worked

Temporal CNN led validation scores

All checkpoints reproduced held-out metrics

Shared features matched live preprocessing

What needs work

Filtering left only 96 swipes

Tracking loss restarted live windows

Stationary replay triggered false shaking

Ablations pending: wrist motion, filtering, class balance

Conclusion & future work

96.8% held-out accuracy

A trained gesture model now powers a local sound prototype.

Reduce stationary-hand false triggers

Improve tracking and swiping recall

Validate webcam, audio, and full-loop delay

Run ablations on existing Jester data

Demo: gestures to sound

C	E	G	A
Roll	Shake	Tap	Swipe

Open localhost:8765

Start camera and show one hand

Repeat each movement for three seconds

Demonstrate mute, then end session

Live prototype demo. Recognition and audio require a pre-presentation check.

Q&A / References

Questions?

Qualcomm: 20BN-Jester gesture dataset

Zhang et al. (2020): MediaPipe Hands

Zhang, Liu & Wu (2019): Mobile action recognition

Inspiration: ericwei97-cloud/gesture-synth

github.com/aditya1nigam8/calmgesture

Development branch: codex/sangini